PRINCIPLES OF DATA SCIENCE · PROJECT REPORT

AirGuard India

A Vulnerability-Weighted Health Advisory Index and Next-Day Air Quality Forecasting System for Indian Cities

7 population cohorts | 12 cities | 18,395 city-days | 3 ML paradigms

COURSE	**AD23532 — Principles of Data Science**
PROGRAMME	B.Tech. Artificial Intelligence and Data Science · Semester V
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LIVE APP	https://webapp-phi-taupe.vercel.app
REPOSITORY	https://github.com/nishaids/airguard-india

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1. What is this project, in one breath?

India already tells you how dirty the air is today. **This project tells you how dirty it will be tomorrow, and it tells seven different kinds of people seven different things — because the same air is not equally dangerous to everyone.**

The simple version

Imagine a school that has one bell for everyone.

When the bell rings, *everybody* goes outside — the football team, the kid with asthma who carries an inhaler, the pregnant teacher, the 70-year-old gardener. One bell, one message, for everyone.

That is exactly what India's official Air Quality Index (AQI) does. It looks at the air over a whole city, turns it into **one number**, and gives that same number to every single person in that city.

But here is the thing: **the same air is not equally harmful to everyone.**

- A healthy 25-year-old footballer breathes dirty air and feels a bit tired.
- A child breathing the *same* air is harmed more, because a child's lungs are still being built and a child breathes more air for each kilogram of body weight.
- Someone with asthma may end up in hospital from that same air.

So one bell is not enough. **AirGuard India rings seven different bells.**

And the second problem: the official AQI tells you about air you have *already breathed*. It is like a weather report that only tells you it rained yesterday. If you want to decide whether your child plays football tomorrow, yesterday's report is useless.

So this project predicts tomorrow.

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2. The two problems, stated properly

PROBLEM	WHAT THE OFFICIAL AQI DOES	WHAT AIRGUARD INDIA DOES
One number for everyone	Reports the maximum pollutant sub-index, identically to every person in a city	Computes a **Health Advisory Index (HAI)** separately for **7 population groups**, each weighted by how much each pollutant actually harms that group
It looks backwards	Summarises air already breathed	**Forecasts the next day** from today's measurements and the recent trend

And the measured result of doing this: on **66.92%** of all 17,057 city-days in the record, **at least one group of people should have been told something different** from what the official AQI implied.

That is not an opinion. It is a count.

3. The 8 major novelties

These are the things that make this a research project and not just a dashboard. Each one comes with the evidence that it actually works — not a claim, a measurement.

★ 1. The Health Advisory Index (HAI) — seven people, seven answers

Simple version: instead of one score for the whole city, we build a separate score for seven kinds of people.

The seven groups:

1. Healthy adults
2. Children (under 14)
3. Elderly (60+)
4. People with asthma or COPD
5. Cardiac (heart) patients
6. Pregnant women
7. Outdoor workers

Each group gets a **vulnerability weight** for each pollutant, based on published medical research. For example, children get a **1.55×** weight on nitrogen dioxide and **1.45×** on ozone, because their airways are still developing and they breathe more air per kilogram of body weight than adults do.

The evidence it matters:

GROUP	AVERAGE HAI	AVERAGE AQI ÷ 5	HOW MUCH HIGHER	GETS A *DIFFERENT* ADVISORY THAN THE AQI
Healthy adult	31.19	29.83	+1.35	**25.95%** of days
Children (under 14)	39.75	29.83	+9.92	**47.05%** of days
Elderly (60+)	43.18	29.83	+13.35	**54.47%** of days
Asthma / COPD	41.12	29.83	+11.29	**50.50%** of days
Cardiac patients	44.88	29.83	+15.05	**58.11%** of days
Pregnant women	41.99	29.83	+12.16	**51.43%** of days
Outdoor workers	42.41	29.83	+12.58	**54.46%** of days

Read the cardiac row again: **on more than half of all days, a heart patient should have been given a different instruction than the one the official index gave them.**

★ 2. The never-less-protective guarantee — a promise proved by code

Simple version: we made a promise, and the computer checks the promise every single time the project runs.

The promise: **the HAI will never tell you the air is safer than the official AQI says.** It can say "more careful", never "less careful". If it ever broke that promise, the whole pipeline would stop with an error.

This matters because a health tool that is *sometimes* less cautious than the official advice is dangerous. So it is not allowed to be, ever — and that is enforced by an assertion in the code, not by hoping.

Three more properties are proved the same way: the index stays inside its bounds, it only goes up when a pollutant goes up (monotonicity), and the ordering between groups is preserved.

★ 3. Forecasting tomorrow, and proving the model actually helps

Simple version: predicting tomorrow's air is easy to *fake*. We made sure we didn't fake it.

Here is the trap. Air quality barely changes overnight. So if you just say **"tomorrow will be exactly like today"** — a rule so dumb it needs no machine learning at all — you already get **$R^2 = 0.841$**, which looks like a great score.

A dishonest project would report "$R^2 = 0.91$!" and let you be impressed.

This project reports the honest number instead:

	MEAN ABSOLUTE ERROR (AQI POINTS)
"Tomorrow = today" (the dumb rule)	18.90
Our LightGBM model	**15.10**
The actual improvement	**20.11% better**

That **20.11%** is the real contribution of the machine learning. Everything above that is free and would have been free without any model.

★ 4. Data leakage demonstrated by experiment, not merely avoided

Simple version: we ran an experiment to *show* how easy it is to cheat, so that nobody can accuse us of cheating.

"Leakage" is when a model secretly gets to peek at the answer. In air quality, there is a very tempting form of it: the AQI *band* (Good / Poor / Severe) is calculated directly from the AQI *number* using a fixed table. So if you ask a model "what band is today?" and also hand it today's AQI number, it will score **100% perfect** — while having learned absolutely nothing. It just read the answer key.

So we fitted three models on identical data splits and published all three:

WHAT THE MODEL WAS ASKED	MACRO-F1 SCORE	WHAT IT MEANS
Today's band, given today's AQI	**1.0000**	Cheating. Reading the answer.
Today's band, with AQI removed	0.7077	The honest same-day version
Tomorrow's band (what we actually do)	**0.7411**	A real forecast

That perfect 1.0 is displayed proudly *as a warning*. It is what our numbers would look like if we were cheating — and they don't look like that.

★ 5. Imputation chosen by measurement — including rejecting the winner

Simple version: some measurements are missing. We tested five ways to fill the gaps, and then deliberately *did not* use the one that scored best.

We hid 15% of the real values on purpose, filled them back in five different ways, and measured which method reconstructed the truth most accurately.

The winner was **linear interpolation** — but linear interpolation fills today's gap by looking at *tomorrow's* value. In a forecasting system that is time travel, and time travel is cheating.

So we rejected the winner and used **forward-fill (limit 3 days)**, which only ever looks backwards. And then we **reported how much accuracy that cost us**, as a number, instead of hiding the trade-off.

★ 6. Event-aware cleaning — Diwali is signal, not noise

Simple version: the dirtiest days of the year are real, not mistakes. We made sure the computer doesn't delete them.

Most data-cleaning code automatically deletes "outliers" — values that look extreme. But in Indian air quality, the extreme days are the *most important* days:

- **Diwali** — firecracker smoke sends pollution through the roof
- **Crop-residue burning season** (October–November in the north)
- **The 2020 COVID lockdown** — when the air suddenly got dramatically cleaner

An automatic outlier filter would erase all three. Ours explicitly **exempts** these known event windows, so genuine pollution episodes are never thrown away as "errors".

The proof it's worth keeping: Diwali produces a measurable national spike, and the lockdown produced a measured **−33.99%** drop in national mean AQI.

★ 7. The safe-exposure window — and catching a subtle data trap

Simple version: we tell you the cleanest hours of the day to go outside — and we had to find the right data to do it, because the obvious data was a trap.

The dataset has hourly AQI values. The obvious thing is to look for the cleanest hour. But there is a hidden problem: **the supplied hourly "AQI" is a 24-hour rolling average.** It is smeared across an entire day by its own definition, so it *cannot* show a daily rhythm. Using it would have shown a pathetic **8%** swing between the best and worst hour and we'd have concluded, wrongly, that the time of day barely matters.

So we rebuilt the calculation from **instantaneous hourly concentrations** instead. The real swing: **39.6%.**

The difference between 8% and 39.6% is the difference between "don't bother" and "a genuinely useful piece of health advice". Both numbers are reported, along with the explanation of why they differ.

★ 8. The counterfactual policy simulator — checked against reality

Simple version: you move a slider that says "cut PM2.5 by 20%", and the project recalculates the *entire* five-year history from scratch, to show how many days would have been safer.

It doesn't just rescale a number. It recomputes every pollutant sub-index, every AQI, every band, and all seven cohort HAI values, for all 18,395 city-days.

Headline scenario — **PM2.5 and PM10 both cut by 20%**:

- Mean AQI: **144.1** → **120.3** (−16.5%)
- **29.8%** fewer unsafe city-days
- **12,432 cohort advisory-days averted**

And here is the part that makes it science rather than a toy: we checked the simulator against something that *actually happened*.

The 2020 national lockdown was a real, sudden, economy-wide emission cut. It produced a **measured −33.99%** change in national mean AQI. The simulator's nearest comparable scenario (all pollutants cut 40%) predicts **−37.0%**.

Those two numbers agree closely — **and the simulator was never fitted to the lockdown in any way.** It had no idea the lockdown existed. That is an independent validation.

Two more, for completeness

The project actually ships **ten** novelties. The two not counted above:

- ★ **City pollution archetypes** — unsupervised clustering on the *shape* of each city's pollutant mix (not how dirty it is), producing two archetypes that map to genuinely different policy levers. Validated by reproducing the exact same grouping with a second, completely different algorithm (**ARI = 1.000**).
- ★ **Explainability and Responsible AI** — TreeSHAP explanations, plus a data card and model card that state every limitation plainly.

> **A finding worth mentioning:** when we asked SHAP which feature the models rely on most, the answer for **both** the regression and the classification model was `HAI_healthy_adult` — **the index this project invented**. It beats raw PM2.5 and even the official AQI itself. We were not aiming for that; it fell out of the analysis, which makes it better evidence.

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4. The dataset — what, from where, and why

What it is

FIELD	VALUE
Name	Air Quality Data in India (2015–2020)
Original source	**Central Pollution Control Board (CPCB)** — Ministry of Environment, Forest and Climate Change, **Government of India**
Where we downloaded it	Kaggle, uploaded by Vopani (`rohanrao`)
Licence	**CC0: Public Domain** — verified live at download time
Raw size	29,531 rows, 26 cities, 2015-01-01 → 2020-07-01
After cleaning	**18,395 rows across 12 cities**
One row means	One city, one day: the average pollution across that city's monitoring stations

Why this dataset

1. **It is official government data.** CPCB is the body that legally operates India's air quality monitoring network. This is not a hobbyist's scrape.
2. **The licence is genuinely open.** CC0 means public domain — usable without restriction. We didn't take that on trust: the licence string was read directly from the live page at download time and stored, along with a **SHA-256 checksum of every file**, in `data/raw/PROVENANCE.json`. Anyone can verify they have byte-identical data.
3. **It is long enough to learn from.** Five and a half years covers five winters, five Diwalis, five monsoons — enough for a model to learn seasonal behaviour rather than memorise one year.
4. **It contains a natural experiment.** The 2020 lockdown gives us a real, independent event to validate the policy simulator against. That is rare and valuable.
5. **It has both daily and hourly files**, which is what makes the safe-exposure window feature possible at all.

Why only 12 of the 26 cities

We set **two rules before looking at any results** (this matters — deciding afterwards would be cherry-picking):

1. At least **80%** of the days in a city's own span must have a real AQI value.
2. The city must have at least **730** labelled days (two years).

14 cities failed. **Every single exclusion is listed with its reason** in the data card — nothing was dropped quietly. Examples:

- **Mumbai** — excluded, only 38.6% complete
- **Patna** — excluded, 78.5% complete (just under the line, still excluded)
- **Aizawl** — excluded, only 111 labelled days

The 12 that survived: Amaravati, Amritsar, Bengaluru, Chennai, Delhi, Gurugram, Hyderabad, Jaipur, Kolkata, Lucknow, Thiruvananthapuram, Visakhapatnam.

What the data does NOT contain (stated honestly)

- **No weather.** No wind, rain, temperature or humidity — and wind is what actually blows pollution away. This is the single biggest limitation and it caps how accurate any model on this data can possibly be.
- **No health outcomes.** No hospital admissions, so the HAI cannot be validated against the illnesses it aims to predict.
- **Urban only**, and the record **ends 1 July 2020** — so it does not describe today's air.

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5. The tech stack

The brain (data science, Python 3.10)

TOOL	JOB
pandas / numpy	Handling the tables and the maths
scikit-learn	Linear/Ridge regression, Random Forest, SVM, KNN, K-Means, metrics
LightGBM	The winning regression model, and the classifier this site serves
XGBoost	The best-scoring classifier
imbalanced-learn (SMOTE)	Creating synthetic examples of rare classes, so "Severe" days aren't ignored
SHAP	Explaining *why* the model predicted what it predicted
matplotlib / seaborn	The 41 figures, all at 300 DPI, all colour-blind-safe
python-docx	Auto-generating the three review documents

The website you are looking at

TOOL	JOB
Next.js 16 + React 19	The website itself, 5 pages
Recharts	The interactive charts
Python serverless functions	The two live prediction endpoints
Vercel	Hosting

Why two different models for classification

The site serves **LightGBM + SMOTE** even though **XGBoost + SMOTE** scored better. Here is the honest reason:

Vercel gives each serverless function a hard **250 MB** limit. We measured the real Linux package sizes rather than guessing:

PACKAGE	INSTALLED SIZE
xgboost	**145 MB**
scipy	105 MB
pandas	64 MB
scikit-learn	45 MB
numpy	40 MB

XGBoost alone eats more than half the budget. Together they came to **320 MB** — over the limit before we'd added a single line of our own code.

What we did about it: we exported the trained model's decision trees to plain JSON and wrote our own ~80-line tree evaluator that needs no machine-learning library at all. Then we verified it reproduces the original model **exactly** — **0.0 difference** across all 4,397 test rows for regression, **0 label mismatches** for classification.

The final bundles: **199 MB** and **168 MB**. Both fit, with room to spare.

The cost of switching classifier is stated openly on the site: macro-F1 **0.7145** (deployed) vs **0.7411** (reported best) — a gap of **−0.0266**.

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6. How the backend works

The simple version

Think of it as a **kitchen** and a **waiter**.

The **kitchen** (the Python pipeline) does all the heavy cooking *in advance* — hours of it. It cleans the data, trains the models, draws the charts, writes the reports. This happens on a laptop, once.

The **waiter** (the website) mostly just carries out food that is already cooked. Almost everything you see on the site was cooked in advance and is just being served from a file.

Only **two** things are cooked fresh to order, because they depend on what *you* click:

1. **Your forecast** — because you choose the city, the date and which group of people you are
2. **Your policy scenario** — because you choose the sliders

The pipeline: 20 stages, one command

```
python run_all.py
```

That runs, in dependency order:

```
acquire → quality → clean → features → safe_windows → eda
  → regression → classification → clustering → tuning → evaluation
  → explain → policy → cards → diagrams → notebooks → reports
  → readme → smoke → audit
```

- **acquire** — download the data, record SHA-256 checksums and the licence
- **clean** — fix missing values, remove sensor errors, protect event days
- **features** — build 100+ predictive columns (yesterday's AQI, 7-day averages, pollutant ratios, all seven HAI values)
- **regression / classification / clustering** — train and compare every model
- **tuning** — 322 model fits searching for better settings (~39 minutes)
- **explain** — compute SHAP values
- **smoke** — 14 tests, including 336 city/cohort/date combinations to prove the app never crashes
- **audit** — check 13 "definition of done" criteria

How a forecast request actually travels

```
You pick "Delhi, 5 Nov 2019, Children"
        |
        ▼
Browser sends POST /api/forecast
        |
        ▼
Vercel wakes a Python function (webapp/api/forecast.py)
        |
        ├─ looks up that city-day's 100 engineered features
        ├─ applies your what-if slider reductions, if any,
        |  and recomputes every affected value from scratch
        ├─ runs the exported regression trees  → predicted AQI
        ├─ runs the exported classification trees → band + confidence
        ├─ computes HAI for all 7 cohorts
        ├─ looks up the cleanest outdoor window
        ├─ checks whether your input looks unlike the training data (drift)
        └─ writes the plain-English advisory
        |
        ▼
JSON comes back; React draws the charts
```

The important design choice: the serverless function does *not* load a `.joblib` model file, because that would require importing LightGBM, which requires scipy, which blows the size limit. It reads the trees from JSON and walks them with plain arithmetic instead.

The one guard-rail worth knowing

The what-if sliders do **not** rewrite history. If you reduce today's PM2.5 by 20%, the code recomputes today's sub-indices, today's AQI and today's seven HAI values — but it leaves yesterday's numbers and the 7-day rolling averages alone.

Why: a what-if about *today* should not retroactively change what happened last Tuesday. Pretending otherwise would make the counterfactual dishonest.

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7. What the website actually does, page by page

Overview

Live headline numbers, the dataset provenance and licence, the two city archetypes, and which models are running.

Forecast — the main tool

Pick a **city**, a **date**, and **which group of people you are**. You get:

- Tomorrow's **predicted AQI number**
- Tomorrow's **predicted band** with the model's **confidence**, plus the full probability spread across all six bands
- Your group's **Health Advisory Index** and risk band
- The **dominant pollutant** that day
- The **cleanest hours** to go outside
- A plain-English **advisory** written for your group
- A **30-day history chart** with the forecast marked, and — for historical dates — what actually happened, so you can see the model's error yourself
- **All seven cohorts side by side**, so you can see how much the advice differs
- A **drift check** that warns you if the day is unlike anything the model was trained on
- **What-if sliders** to cut pollutants and watch everything recalculate

City Risk Atlas

Five tabs: the two discovered **archetypes**, a **ranked risk table** of all 12 cities, a **month-by-city heatmap**, the **Diwali** analysis, and the **2020 lockdown** analysis.

Policy Simulator

Move sliders (or pick a preset), and the entire five-year record is re-scored. Shows AQI change, unsafe days removed, band migration, per-city effect, and cohort advisory-days averted — plus the lockdown sanity check.

Model Insights

Seven tabs of full evidence: every model compared, confusion matrices, ROC curves, residual diagnostics, SHAP explanations, the tuning report, the leakage demonstration, a plain-language metric glossary, and the data and model cards.

8. The final scoreboard

TASK	MODEL	SCORE	BEAT THE DUMB BASELINE BY
Predict tomorrow's AQI number	LightGBM	MAE **15.10**	**+20.11%**
Predict tomorrow's band	XGBoost + SMOTE	Macro-F1 **0.7411**	**+8.94%**
Group cities by pollution type	K-Means (k=2)	Silhouette **0.3553**	ARI **1.000** vs a second algorithm

The safety number that matters most: of 4,397 predictions, only **5 (0.114%)** understated the danger by two bands or more. That is the failure that could actually hurt somebody — being told the air is cleaner than it is — so it is reported as a headline number instead of being buried in an average.

Also: 41 figures at 300 DPI · 14 smoke tests over 336 combinations, zero crashes · 12 of 13 definition-of-done checks passing · every number in every report read from JSON at generation time, never typed in by hand.

9. The honest limitations

A good project states what it cannot do:

1. **No weather data** — and wind is the single biggest driver of whether pollution disperses. This caps achievable accuracy.
2. **No health outcomes** — the HAI is grounded in published exposure-response literature, but it has not been validated against actual hospital admissions, because this dataset has none.
3. **The record ends 1 July 2020** — nothing here describes today's air, and the final months are atypical because of the lockdown.
4. **12 cities, urban only** — no rural monitoring, though much of India's biomass-burning exposure is rural.
5. **One value per city per day** — cannot tell a roadside from a park.
6. **This is not medical advice.** It is general public-health guidance. The official CPCB index remains the authoritative index for India.